Variation in Penguin Body Mass Across Islands in the Palmer Archipelago

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Abstract

Body mass is an important biological trait that can vary across populations due to ecological and environmental factors. In penguins, variation in body mass may reflect differences in habitat conditions, resource availability, or energetic demands across locations. This study examined whether penguin body mass differs across three islands in the Palmer Archipelago using the Palmer Penguins dataset. Body mass measurements from penguins sampled on Biscoe, Dream, and Torgersen islands were analyzed using a one-way analysis of variance (ANOVA). Mean body mass differed significantly among islands, and a Tukey's HSD test was applied to identify pairwise differences. Penguins on Biscoe Island exhibited significantly greater body mass than those on Dream and Torgersen islands, while no significant difference was observed between Dream and Torgersen. These findings suggest spatial variation in penguin body mass across the Palmer Archipelago.

Introduction

Body size is a fundamental biological trait that influences physiology, survival, and reproductive success across animal taxa. In ecological systems, variation in body size is often associated with environmental gradients, including temperature, resource availability, and habitat structure. One well-known ecological pattern, Bergmann's rule, predicts that individuals in colder environments tend to exhibit larger body sizes, potentially due to thermoregulatory advantages (Blackburn et al., 1999).

Penguins (genus Pygoscelis) provide a useful system for examining variation in body mass across spatial environments. These seabirds inhabit diverse ecological niches within the Antarctic region, where differences in ocean productivity, prey availability, and environmental conditions may influence individual growth and energetic balance. Previous studies have demonstrated that penguin morphology can vary with environmental conditions and ecological pressures, including differences in foraging strategies and reproductive investment (Gorman et al., 2014).

The Palmer Archipelago, located along the western Antarctic Peninsula, contains multiple islands that support penguin populations under varying local conditions. The Palmer Penguins dataset provides standardized morphological measurements collected across these islands, offering an opportunity to examine spatial variation in traits such as body mass (Horst et al., 2022).

In this study, we test whether penguin body mass differs across three islands in the Palmer Archipelago (Biscoe, Dream, and Torgersen). We hypothesize that mean body mass will vary among islands, reflecting differences in local environmental conditions and ecological factors.

Methods

Data Source

Data were obtained from the palmerpenguins R package (Horst et al., 2022). The raw dataset was exported to CSV using scripts/01_load_raw_penguins.R.

Data cleaning was performed using scripts/02_clean_penguins.R, which:

-Selected the variables island and body_mass_g -Removed rows containing missing values -Generated a processed dataset for analysis

Statistical Analysis

All analyses were performed in R. Differences in mean penguin body mass across islands were evaluated using a one-way analysis of variance (ANOVA), with island treated as a categorical predictor and body mass as the response variable. Where a significant overall effect was detected, Tukey's Honestly Significant Difference (HSD) test was used for post-hoc pairwise comparisons between islands. Statistical significance was assessed at $\alpha = 0.05$.

Results

Summary statistics for penguin body mass across islands are presented in Table 1.

Table 1: Summary statistics for penguin body mass across islands.

	Island	Mean	SD	N
Biscoe	Biscoe	4716.0	782.9	167
Dream	Dream	3712.9	416.6	124
Torgersen	Torgersen	3706.4	445.1	51

To visualize the distribution of body mass across islands, a boxplot was generated (Figure 1). Penguins sampled from Biscoe Island appear to have higher body mass values overall compared with those from Dream and Torgersen islands.

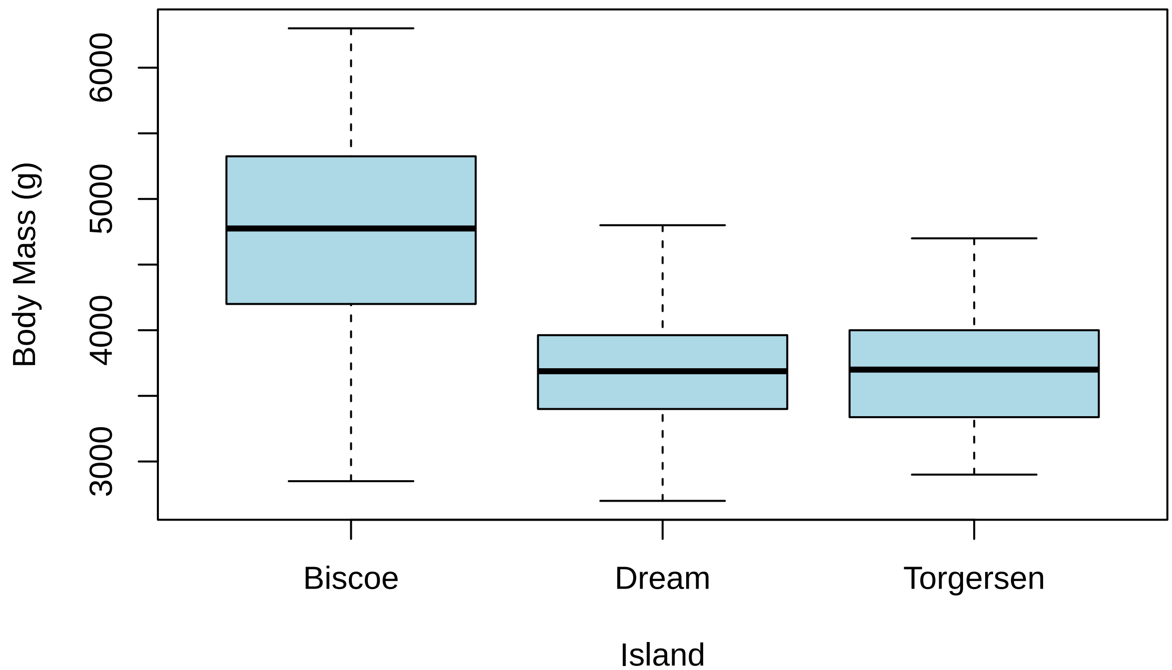

Figure 1: Distribution of penguin body mass across the three islands.

A one-way analysis of variance (ANOVA) is presented in Table 2 and revealed a significant effect of island on body mass ($F(2, 339) = 110$, $p < 0.001$). Post-hoc comparisons using Tukey's Honestly Significant Difference (HSD) test indicated that penguins from Biscoe Island had significantly greater body mass than those from Dream and Torgersen islands ($p < 0.001$ for both comparisons), while body mass did not differ significantly between Dream and Torgersen islands ($p = 0.998$).

Table 2: ANOVA results for differences in penguin body mass across islands.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Island	2	86314512	43157255.8	110.01	0
Residuals	339	132993186	392310.3	NA	NA

Post-hoc comparisons using Tukey's Honestly Significant Difference (HSD) test indicated that penguins from Biscoe Island had significantly greater body mass than those from Dream and Torgersen islands ($p < 0.001$ for both comparisons), while body mass did not differ significantly between Dream and Torgersen islands ($p = 0.998$).

Discussion

This study demonstrates that penguin body mass varies significantly across islands within the Palmer Archipelago. Penguins sampled on Biscoe Island exhibited substantially greater body mass compared to those on Dream and Torgersen islands, while the latter two islands did not differ significantly from one another. These results suggest that spatial variation in environmental conditions may influence penguin body mass at the population level.

Differences in body mass across islands may reflect variation in ecological factors such as prey availability, ocean productivity, or energetic demands associated with local environmental conditions. Larger body mass in penguins from Biscoe Island may indicate access to greater or more consistent food resources, or differences in habitat conditions that support increased energy storage. Alternatively, variation in species composition across islands may also contribute to observed differences in body mass, as different penguin species exhibit distinct size ranges.

It is important to note that this analysis is based on observational data and does not account for potential confounding factors such as species identity, sex, or age. As a result, causal interpretations cannot be made. Future analyses incorporating additional variables could provide a more detailed understanding of the mechanisms underlying variation in penguin body mass.

Despite these limitations, this study highlights clear spatial differences in penguin body mass across islands in the Palmer Archipelago and demonstrates the utility of reproducible workflows for conducting and communicating ecological analyses.

Because this dataset is observational, results should not be interpreted as causal. Future research should take species categorization into consideration when designing the study.

Data and Code Availability

All data processing and analysis scripts are available in the scripts/ directory. The processed dataset is available in data/data_processed/, and all outputs are stored in the outputs/ directory.

References

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